

Behaving Sustainability: Cultural Cognition, Environmental Orientation, and Pro-Environmental Behavior

Introduction

Achieving environmentally sustainable outcomes requires some behavioral changes at the individual, community, and societal levels. However, not all individuals are equally likely to engage in pro-environmental behaviors, and understanding the psychological and cultural factors that drive these behaviors is important for enhancing sustainability. Scholars of environmental psychology often lament the so-called ‘attitude-behavior’ gap wherein environmental attitudes do not explain very much of the variance in environmental behavior (Gifford and Nilsson 2014; Kollmuss and Agyeman 2002; Stern 2000). If people care strongly about the environment, why don’t they act in a more sustainable way? The attitude-behavior gap is not because people are hypocrites, but because researchers do not yet have a complete attitude-behavior model (Kaiser, Kibbe, and Hentschke 2021).

To understand how attitudes influence behaviors, scholars use approaches such as the Value-Belief-Norm (VBN) framework, where higher-order values shape second-order beliefs that, in turn, influence behaviors []. Higher-order values form a cognitive foundation that influences how information about the world is processed and understood across multiple issues []. When considering environmental behaviors, the second-order values involve environmental cognition, such as views about the relationship of humans to the natural world and how information about the environment is understood []. In addition to environmental cognition, there are also affective feelings that individuals hold about the environment []. Together environmental cognition and affective feelings form an environmental orientation that shapes environmental behaviors []. However, to develop a complete attitude-behavior model, it is important to understand all of the various links in the chain, including the higher-order values and cognition that inform environmental orientations. Of note, the cultural worldviews measured by cultural cognition have been shown to shape environmental orientations (Nowlin 2019).

Cultural cognition was developed by Kahan and others (Kahan et al. 2007) as a tool to measure how cultural values shape beliefs about risk. It is based on cultural theory and its conceptualization of worldviews []. Cultural theory addresses how cultural values influence how people perceive and respond to risk. By measuring the cultural values of an individual, we may be able to predict how they will perceive and respond to risks

associated with certain activities or products. We are interested in how cultural cognition shapes views toward environmental issues and how those views shape behavior.

The cultural worldviews associated with cultural cognition are measured in two dimensions, a grid dimension (egalitarianism-hierarchy) and a group dimension (communitarianism-individualism). In general, those that are more egalitarian and communitarian are more concerned about environmental risks than those that are more hierarchical and individualist. Therefore, it is expected that individuals high in egalitarianism and communitarianism should engage in pro-environmental behaviors to mitigate risks to the natural environment. However, previous research shows only a weak correlation between egalitarianism and communitarianism and pro-environmental behavior (Sparks et al. 2021). Therefore, cultural cognition shapes how individuals think and process information about issues, but may not directly influence behaviors.

In this paper, we develop a theoretical framework that accounts for both the cognitive and affective influences on pro-environmental behavior. Specifically, we posit that higher-order beliefs, drawn from cultural cognition (Kahan 2012; Kahan et al. 2012; Kahan and Braman 2006), inform environmental orientations (Henry and Dietz 2012), which in turn influence behavior, both public and private. Drawing from Henry and Dietz (2012), we use the New Ecological Paradigm (NEP) as our measure of environmental cognition, and the Connectedness to Nature Scale (CNS) as the affective element of environmental orientation. Previous work failed to find that cultural cognition influence environmental behavior; however, it has been found to influence environmental cognition using the NEP (Nowlin 2019). Therefore, we hypothesize that environmental orientations mediate the relationship between cultural cognition and behavior.

Overall we find ...

Next, we develop our approach based on previous literature.

Cognition, Environmental Orientations, and Pro-Environmental Behavior

The process through values and beliefs impact behavior is often understood as a process where higher-order beliefs influence second-order beliefs, which in turn influence behavior. One of the leading frameworks that adopts this approach is the Value-Belief-Norm (VBN) framework (Stern et al. 1999). In brief, the VBN posits that underlining values inform broad environmental orientations that activate norms that in turn support pro-environmental behavior. VBN involves a sequential chain that goes from values to beliefs to behavior. Therefore, values inform attitudes, which in turn inform behaviors.

A similar set of assumptions inform a sophisticated understanding of belief systems developed by policy scholar using the Advocacy Coalition Framework (ACF) (Henry and Dietz 2012; Nohrstedt et al. 2023; Sabatier and Jenkins-Smith 1993). With this approach, beliefs systems are made up of three different types of beliefs, including deep core, policy core, and secondary beliefs. Deep core beliefs are the broadest in scope and are fundamental beliefs that inform attitudes about multiple issues. Political ideology is an example of a deep core belief. Next, are policy core beliefs, which are beliefs about particular issues. Environmental orientations would be akin to policy core beliefs. Finally, secondary aspects are the narrowest in scope and include beliefs about how best to achieve the goals outlined in policy core beliefs.

In developing our approach, we draw on the notion of a sequential chain and a hierarchical ordering of values, beliefs, and attitudes that lead from abstract values to concrete actions. Specifically, we note that deep core beliefs inform environmental orientations that, in turn, make pro-environmental behavior more (or less) likely. In addition, we separate beliefs that shape *cognition* from the affective and identity elements of environmental orientations. Our measure of deep core beliefs is cultural cognition, which as the name implies, are deep rooted values that influence cognition. For environmental orientations, we draw on environmental cognition, which is “the way individuals structure their thinking about environmental issues and associated political actions” (Henry and Dietz 2012, 238). We argue that environmental orientations contain a cognitive element as well as an affective element. Deep core beliefs influence both the cognitive and affective aspects of environmental orientation. Finally, deep core beliefs and environmental orientations influence pro-environmental behaviors; however, deep core beliefs likely influence behaviors *indirectly* (Sparks et al. 2021), with the affective element of environmental orientations having the most direct influence on behaviors [].

Deep Core Beliefs: Cultural Cognition

For our measure of deep core beliefs we use cultural cognition. Cultural cognition is derived from the cultural theory of risk developed by Mary Douglas and Aaron Wildavsky (Douglas and Wildavsky 1982; Thompson, Ellis, and Wildavsky 1990). Cultural theory (CT) has been identified as a leading way to conceptualize and measure deep core beliefs (Ripberger et al. 2014). In brief, CT posits that social structures and relationships are represented by two dimensions, including a grid dimension and a group dimension. Four competing cultural worldviews arise based on their position on the grid and group dimensions. The grid dimension represents preferred social structures regarding the roles that one plays in society and the rules by which one must abide. The group dimension represents the importance of cohesive social units and the well-being of the broader community. The original formulation of CT produces four cultural worldviews based on the relative positions on the grid and group dimensions, including hierarchy (high grid, high group), egalitarianism (low grid, high group), individualism (low grid, and low group), and fatalism (high grid, low group).

Cultural worldviews provide a foundational set of deep core beliefs that inform beliefs about specific policy issues (i.e., policy core beliefs), societal controversies, and risks, among other beliefs (Jenkins-Smith et al. 2014; Nowlin 2019; Ripberger et al. 2014; Sotirov and Winkel 2016). In the context of cultural worldviews, perceptions of risk are based on the degree to which they undermine or threaten the preferred social order within each worldview. Several studies have shown a connection between cultural worldviews and risk perceptions associated with a host of environmental issues (see Xue et al. 2014), including fracking (Boudet et al. 2014), nuclear waste (Jenkins-Smith and Smith 1994), renewable energy (Tumlison et al. 2018), geoengineering (Kahan et al. 2015), high-voltage power lines (Moyer and Song 2016, 2019), and climate change (Jones 2011; Kahan et al. 2012; Nowlin 2022; Nowlin and Rabovsky 2020; Price, Walker, and Boschetti 2014), among others.

Cultural cognition draws CT and its cultural worldviews, but makes two important distinctions. The first is its focus on social and psychological mechanisms that examine how cultural worldview’s shape interpret and process information, understand risks, and take positions on policy issues. Kahan (2012) identifies five mechanisms, including *cultural identity-protective cognition*, where individuals conform their beliefs to those in their “group” that share cultural values; *culturally biased assimilation and cultural polarization*, involves selective attention to worldview-confirming information and balanced information tends drives people of

opposing worldviews further apart rather than together; *cultural availability* occurs as worldview-confirming events and experts are more readily noticed, credited, and recalled; *cultural credibility heuristic* is when people trust sources whose apparent values match their own; and *cultural-identity affirmation*, has been shown when framing information alongside an identity-affirming rather than identity-threatening policy solution makes people more receptive to the underlying facts. Together, these mechanisms account for the cognition part of cultural cognition.

The second distinction of cultural cognition from CT is how cultural worldviews are measured empirically. As noted, the CT worldviews are based on the placement on the grid and group dimensions, and worldviews are a result of placement on both. However, cultural cognition conceives grid and group as two orthogonal dimensions with worldviews identified as being high or low on each dimension rather than a combination of the dimensions. Those high on the grid dimension are called hierarchical as they prefer a society with well-defined roles and clear lines of authority, whereas those low on the grid dimension are termed egalitarian as they value a social order based on an equal sharing of power and resources. Communitarian is the worldview that is high on the group dimension, and it represents a preference for a collectivist society. Finally, individualism is low on the group dimension preferring increased individual autonomy and responsibility (Kahan 2012; Kahan et al. 2012; Kahan and Braman 2006).

Regarding worldviews and environmental issues, in general, communitarianism, with its concern about collective well-being and egalitarianism, with its concern about the negative impacts of concentrated power tend to see threats to the natural world as a large risk. Hierarchs believe that natural resources can be well managed by experts and individualism supports the use of natural resources in private enterprise, and therefore hierarchy and individualism tend not to be concerned about environment harm. As noted, cultural worldviews have been found to be associated with attitudes about environmental issues, yet not associated with pro-environmental behaviors (Sparks et al. 2021). However, cultural cognition is associated with pro-environmental orientation, specifically the New Ecological Paradigm (NEP) (Nowlin 2019).

New Ecological Paradigm

To understand environmental behavior, scholars often turn to mental attributes related to environmental orientations. Our first measure of environmental orientations is the New Ecological Paradigm (NEP). The NEP, developed in 1978 (Dunlap and Van Liere 1978), has been widely adopted to explain environmental behavior (e.g., Stern 2000). Its original development focused on explaining the values associated with environmentalism as an emergent environmental paradigm that challenged the dominant social paradigm. Similar to cultural cognition, the NEP provides an ecological worldview that encompasses values associated with the environment, and, as a worldview it includes cognition elements that shape how the world is seen []. Additionally, it is often used in the VBN as a measure of environmental beliefs (Canlas, Karpudewan, and Khan 2022). The NEP was updated in 2000, and the new measure comprises five aspects which are “the reality of the limits to growth, anti-anthropocentrism, the fragility of earth’s balance, rejection of exemptionalism, and the possibility of an ecocrisis” (Dunlap et al. 2000, 432).

The NEP is not without skeptics. One study applying scale analysis of the NEP based on a random sample of Norwegians found that the scale is not unidimensional as is often claimed (Grendstad 1999). In another

study, Sparks, Ehret, and Brick (2022) criticize the NEP for failing to adequately identify political conservatives who also hold a pro-environment orientation. Most important for our purposes is a recent study that found the NEP to be the weakest predictor of self-reported pro-environmental behavior, accounting for 12% of the variation, when compared in the same sample to the Connectedness to Nature Scale and environmentalist identity (Sparks et al. 2021). This may be because endorsement of the NEP has decreased over time (Lou et al. 2022), or it may be that feeling close and connected to nature is more proximate to behavior than more abstract, cognitive worldviews like cultural cognition and the NEP.

Connectedness to Nature

A second common instrument measuring environmental orientation is the Connectedness to Nature Scale (CNS) (Mayer and Frantz 2004). As implied by its name, the CNS seeks to measure an individual's emotional connection to nature, rather than an environmental worldview or attitude. Mayer and Franz draw inspiration from Leopold's insistence that to address environmental problems, people must feel emotionally connected to the non-human community on which we are dependent upon. The idea is that current ecological crises are a direct result of humanity losing touch with the broader community in which we live. The hypothesis can be tested by measuring an individual's affective connection to nature and determining if someone with more emotional connection correlates to more environmentally protective behaviors. They demonstrate this in their study and show that the CNS better predicts behavior (Mayer and Frantz 2004).

Going from scale development, to specific applications, one study assessed the role of personality traits on environmental behavior (Brick and Lewis 2016). The study also provided a comparison of the mediation effect of both the NEP and the CNS on environmental behaviors. The personality traits Openness and Conscientiousness were both shown to be positive predictors of environmental behavior. In both cases, the CNS had a stronger mediation effect than NEP. In other words, the CNS better predicts pro-environmental outcomes. However, these authors also note that the CNS contains language that is more consistent with liberal environmentalism, and so may fail to adequately capture the full dimensionality of environmental support across the political spectrum (Brick and Lewis 2016).

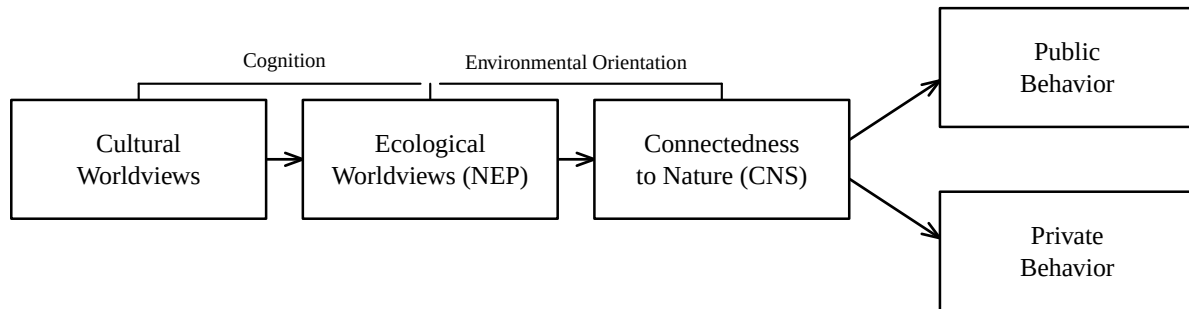
Like the NEP, the CNS is composed mostly of items that tap into politically progressive environmentalism and thus may miss conservative pro-environmental orientation (Sparks, Ehret, and Brick 2022). However, the CNS is a much stronger predictor of environmental behavior – a smaller attitude-behavior gap – than the NEP, the CNS plus controls accounting for 35% of the variance in environmental behaviors (Sparks et al. 2021).

Conceptual Model and Hypotheses

Cultural cognition can be thought of as a deep core belief, and therefore it is an antecedent to environmental orientations. Cultural-based views of risk and social-psychological tendencies likely precede the development of environment-specific cognition, attitudes, and orientations. And given the weak correlation between cultural cognition and environmental behavior it is reasonable to view this as an indirect relationship.

Taking the step from cultural worldviews to environmental orientations, we posit that environmental orientations contain both a cognitive, worldview element that can be measured by the New Ecological Paradigm (NEP) and an affective, emotional connection to nature, as measured by the Connectedness to Nature Scale (CNS). Additionally, we posit that emotional connections to nature are most predictive of pro-environmental behaviors than cultural or ecological worldviews. As noted, cultural worldviews are an antecedent to environmental orientations, and similarly, we posit that ecological worldviews are an antecedent to connectedness to nature. Finally, connectedness to nature influences both public and private pro-environmental paper. Figure 1 illustrates the conceptual model connecting cognition, based on cultural and ecological worldviews; environmental orientations, including both the ecological worldview component and the affective connection to nature component; with both public and private pro-environmental behaviors.

Figure 1: Conceptual model tested with the structural equation model: cultural cognition shapes environmental orientation—first the New Ecological Paradigm (NEP) and then the Connectedness to Nature Scale (CNS)—which in turn shapes public and private pro-environmental behavior.



Based on the model shown in Figure 1, we posit several hypotheses regarding the relationships between cultural cognition, the NEP, the CNS, and environmental behavior. As noted, egalitarianism and communitarianism has been shown to be predictive of concern regarding several environmental issues and environmental orientations. Therefore, we expect that:

H1: Egalitarianism positively correlates with the ecological worldview element of environmental orientation

H2: Communitarianism positively correlates with environmental behavior H2b: Communitarianism positively correlates with environmental orientation (NEP and CNS)

Previous research indicates that cultural cognition predicates environmental orientation and environmental orientation predicts pro-environmental behavior. Therefore, we posit that:

H3: Environmental orientation positively mediates the relationship between egalitarianism and environmental behavior

H4: Environmental orientation positively mediates the relationship between communitarianism and environmental behavior

Our hypotheses are regarding a sequence approach, where cultural worldviews influence ecological worldviews, ecological worldviews influence the connectedness to nature, and connectedness to nature influence both public and private pro-environmental behaviors. However, in our analysis we also examine the direct

influences of cultural worldviews on the CNS and behaviors as well as the direct influence of the NEP on behaviors. Additionally, through mediation analysis, we examine the direct, indirect, and total effects of cultural worldviews and the mediating effects of environmental orientations on environmental behavior.

Data and Methods

To examine the relationship between cultural cognition, environmental orientations, and behavior we draw on a sample of 501 adults living in the US purchased through Survey Sampling Inc. The survey was administered online in July 2017. The sample size was determined by cost considerations. IRB approval for the survey was approved by [redacted for blind review] Human Subjects Committee on April 3, 2017, protocol number 37-17-0256. Our analysis was not pre-registered.

Measures

Our dependent variable is a 12-item scale of self-reported pro-environmental behaviors with the questions taken from the 2000 Gallup Earth Day Poll. Respondents were given a list of several environmental behaviors and asked, “Which of these, if any, have you, yourself, done in the past year?” Possible responses included, “Yes, have done; No, have not done; No opinion.” Yes, responses were coded as a 1 and no/no opinion were coded as 0. For the purposes of analysis, we treat the behavior measure as observed, and we add the number of self-reported behaviors together to produce a measure that ranges from 0 (no behaviors) to 13 (all of the behaviors). The mean number of self-reported behaviors was 5.52, with a standard deviation of 2.98. Table 1 shows the wording for each behavior as well as the mean and sd for each self-reported behavior.

Table 1: Self-reported pro-environmental behaviors by type (public/private), with means and standard deviations

Behavior	Type	Mean	SD
Been active in a group/organization that works to protect the environment	Public	0.19	0.39
Voted for or worked for candidates because of their environmental positions	Public	0.40	0.49
Contributed money to an environmental, conservation, or wildlife group	Public	0.33	0.47
Contacted a public official about an environmental issue	Public	0.19	0.39
Contacted a business to complain about environmentally harmful products/policies	Public	0.16	0.36
Signed a petition supporting environmental protection	Public	0.36	0.48
Attended a meeting concerning the environment	Public	0.17	0.37
Public behaviors (0-7)		1.80	1.92
Avoided using products that harm the environment	Private	0.66	0.47
Tried to use less water in your household	Private	0.75	0.43
Bought a product because it was better for the environment	Private	0.62	0.49
Voluntarily recycled newspapers, glass, aluminum, motor oil, etc.	Private	0.79	0.41
Reduced your household’s use of energy	Private	0.78	0.42

Table 1: Self-reported pro-environmental behaviors by type (public/private), with means and standard deviations

Behavior	Type	Mean	SD
Private behaviors (0-5)		3.60	1.58
Total behaviors (0-12)		5.40	2.86

As shown in Table 1, recycling had the highest mean among the self-reported behaviors at 0.79, which indicates that 79% of the respondents reported that they recycle. Recycling was followed closely by reducing household energy use at 0.78 and using less water at home at 0.75. The behaviors with the lowest means included buying/selling stocks at 0.12, contacting a business at 0.16, and attending a meeting at 0.17.

Cultural Cognition

To measure cultural cognition, we rely on the short form of the cultural cognition survey questions which includes three questions for each of the four cultural worldviews (see Kahan 2012). Respondents were asked their agreement on a 1 (strongly disagree) to 4 (strongly agree) scale with each of the following statements. The hierarchy statements included, 1) We have gone too far in pushing equal rights in this country, 2) It seems like blacks, women, homosexuals and other groups don't want equal rights, they want special rights just for them, and 3) The women's rights movement has gone too far. The egalitarianism statements are, 1) We need to dramatically reduce inequalities between the rich and poor, whites and people of color, and men and women, 2) Our society would be better off if the distribution of wealth was more equal, and 3) We live in a sexist society that is fundamentally set up to discriminate against women. The individualism statements included, 1) If the government spent less time trying to fix everyone's problems, we'd all be a lot better off, 2) Government regulations are almost always a waste of everyone's time and money, and 3) Our government tries to do too many things for too many people. We should just let people take care of themselves. Finally, the communitarianism statements are, 1) Sometimes government needs to make laws that keep people from hurting themselves, 2) Government should put limits on the choices individuals can make so they don't get in the way of what's good for society, and 3) The government should do more to advance society's goals, even if that means limiting the freedom and choices of individuals.

New Ecological Paradigm

To measure the New Ecological Paradigm (NEP) we used agreement with the following three statements on a 1 to 5 scale (1 - strongly disagree, 3 - neutral, 5 - strongly agree). The statements are, 1) Humans are severely abusing the environment, 2) The so-called "ecological crisis" facing humankind has been greatly exaggerated (reverse-coded), and 3) If things continue on their present course, we will soon experience a major ecological catastrophe.

Connectedness to Nature

We measured the Connectedness to Nature Scale (CNS) using respondent agreement on a 1 to 5 scale (1 - strongly disagree, 3 - neutral, 5 - strongly agree) with three statements. The statements included, 1) I think of the natural world as a community to which I belong, 2) I often feel a kinship with animals and plants, and 3) I feel as though I belong to the Earth as equally as it belongs to me.

Table 2 shows the measurement variables used for the latent constructs including the mean and standard deviation (sd) of each statement. In addition, Table 2 show the fit statistics for each of the latent constructs including Cronbach's α , Composite Reliability (CR), and the Average Variance Extracted (AVE).

Table 2: Measurement items and construct reliability (Cronbach's α , composite reliability, average variance extracted)

Construct	Item	Mean	SD	α	CR	AVE
Hierarchy	cc_eh_2	2.33	1.01	0.83	0.83	0.62
	cc_eh_5	2.43	1.08			
	cc_eh_7	2.16	0.98			
Egalitarianism	cc_eh_11	3.13	0.88	0.77	0.79	0.55
	cc_eh_13	2.94	0.91			
	cc_eh_14	2.60	0.87			
Individualism	cc_ci_2	2.69	0.91	0.80	0.80	0.57
	cc_ci_3	2.53	0.90			
	cc_ci_12	2.58	0.86			
Communitarianism	cc_ci_13	3.01	0.70	0.61	0.64	0.38
	cc_ci_14	2.44	0.83			
	cc_ci_16	2.31	0.85			
CNS	cns2	3.78	0.90	0.76	0.72	0.47
	cns6	3.69	1.05			
	cns7	3.71	1.00			
NEP	nep5	3.99	1.14	0.70	0.73	0.48
	nep10	3.25	1.38			
	nep15	3.80	1.15			

As shown in Table 2, each of the constructs show sufficient reliability (α and CR > 0.70) except for the communitarianism scale. However, CR values between 0.60 and 0.70 are acceptable for exploratory analysis (Hair et al. 2015).

To test our hypothesis, we estimated a Structural Equation Modeling (SEM) using the lavaan package in R (Rosseel 2012). The dependent variable, self-reported pro-environmental behaviors, is an observed variable and the cultural cognition worldviews, the CNS, and the NEP scale are latent constructs. We run separate analyses for the two measures of environmental orientation, CNS and NEP, and for both models we show the standardized regression coefficients for the measurement and structural models. In addition, we assess the fit

of both models using the Root Mean Square Error of Approximation (RMSEA), the Comparative Fit Index (CFI), and the Standardized Root Mean Square Residual (SRMR). Finally, we examine the mediation effects of environmental orientation by examining the direct, indirect, and total effects of cultural cognition measures on pro-environmental behavior.

Results

Table 3: Fit comparison of the strict serial chain versus the full combined model with direct paths added. Both models control for age, gender, race, education, and household income.

Model	Chi-sq	df	CFI	TLI	RM-SEA	RMSEA 90% CI	SRMR	AIC	BIC
Serial chain (HI/EC)	199.70	52	0.877	0.801	0.075	[0.064, 0.086]	0.048	11961.1	12129.7
Full (HI/EC, + direct paths)	151.85	44	0.910	0.829	0.070	[0.058, 0.082]	0.039	11929.2	12131.6

Nested χ^2 difference test (serial chain nested in full model, HIER-INDIV/EGAL-COMM groups): $\Delta\chi^2(8) = 47.85, p = 0.0000$.

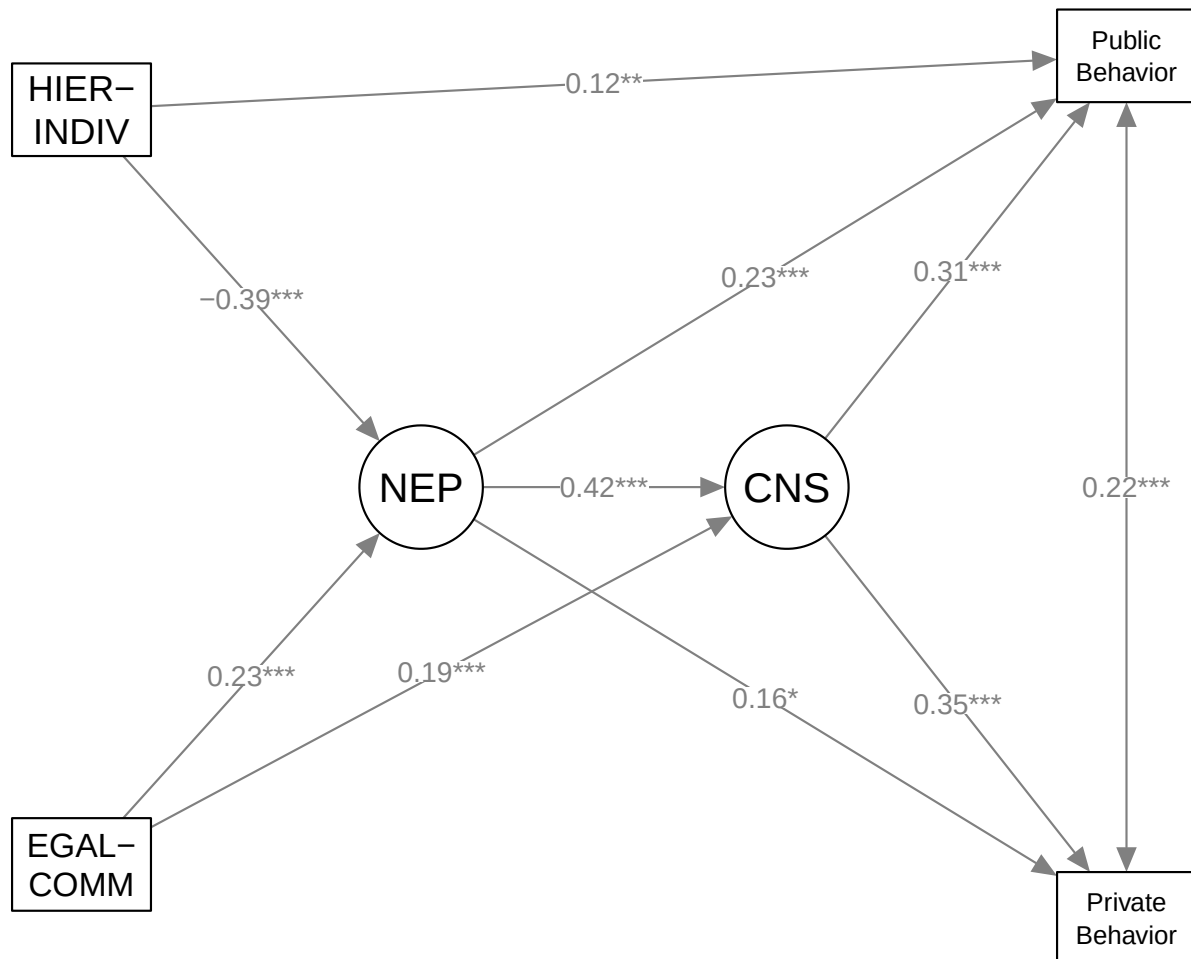
Table 4: Standardized decomposition of cultural worldview group effects on public and private behavior in the full combined model, controlling for age, gender, race, education, and household income ($\dagger p < 0.10$, $*$ $p < 0.05$, $** p < 0.01$, $*** p < 0.001$)

Predictor	Out-come	Direct	Serial (NEP→CNS)	Total indirect	Total
Hierarchy-Individualist	Public	0.122**	-0.052***	-0.138***	-0.016
Hierarchy-Individualist	Private	-0.085†	-0.058***	-0.116***	-0.201***
Egalitarian-Communitarian	Public	0.073†	0.031**	0.142***	0.216***
Egalitarian-Communitarian	Private	-0.041	0.035***	0.137***	0.096*

Discussion

Scholars have noted a gap between attitudes and beliefs about the environment and pro-environmental behavior. Additionally, cultural worldviews derived from cultural cognition, specifically egalitarianism and communitarianism, have been shown to inform views about several environment issues as well as overall attitude toward the environment. But previous research has not found a connection between cultural worldviews and environmental behavior. In this paper, we developed and tested a conceptual model, based on previous research, where cultural worldviews influence environmental orientations and behaviors, and environmental orientations

Figure 2: Structural model of the full combined SEM: cultural worldview groups (HIER-INDIV, EGAL-COMM) → New Ecological Paradigm → Connectedness to Nature → public and private pro-environmental behavior, with direct paths. Controlling for age, gender, race, education, and household income (paths not shown). The measurement (indicator) portion is omitted for clarity; standardized coefficients shown; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The full path diagram including the measurement model, and the equivalent model with the four separate worldview measures, are in the supplemental materials.



influence behaviors. Additionally, we examined the mediation effects of environmental orientations on the relationship between cultural cognition and pro-environmental behaviors. Environmental orientations were measured in terms of attitudes, using the NEP, and the emotional connection to nature, using the CNS. We used SEM to test each of our hypotheses.

We posited that egalitarianism and communitarianism would be positively associated with environmental orientations and environmental behavior (H1 a and b, H2 a and b). We found mixed results for our hypotheses. As expected, egalitarianism is associated with both the CNS and NEP; however, egalitarianism was only associated with behavior in the CNS model. However, communitarianism was associated with environmental orientation only in the CNS model and not associated with behaviors in either model.

Additionally, we posited that environmental orientations mediate the relationships between egalitarianism (H3) and communitarianism (H4) and pro-environmental behaviors. As expected, both the NEP and the CNS significantly mediate the relationship between both egalitarianism and communitarianism, and environmentally protective behavior. As far as we know, this is a novel finding. While previous studies have shown a relationship between cultural cognition and environmental orientation (Nowlin 2019) none have included a mediation analysis. Along with the results of the structural equation modeling, this helps us understand cultural cognition as an antecedent of environmental orientation and fill some of the attitude-behavior gap.

We also found that the Connectedness to Nature Scale likely had more predictive power regarding environmental behaviors than the NEP based on the model fit statistics in Table 3. This finding is in line with work showing that the CNS is a stronger predictor than the NEP (Sparks et al. 2021).

There are at least two potential reasons for the CNS displaying more power than the NEP to influence behavior, which we are only able to speculate about in the present study. First, prior work has demonstrated a slow decline in environmental concern as measured by the NEP (Lou et al. 2022). Lou and colleagues do not provide evidence for why this may be happening, only show through meta-analysis this downward trend. We are not aware of a similar study tracking CNS over time, but that may prove a fruitful avenue for further study. If it is only the NEP that is declining over time, that may indicate that the measurement instrument is becoming outdated, potentially because of shifts over time making the distinction between the dominant social paradigm and the new ecological paradigm less apt. For example, many see technological innovation as the key to unlocking a new green economy, rather than destroying the earth. Relatedly, perhaps the language used in the NEP has just become outmoded and out of line with how environmentalists currently would express environmental attitude (see Sparks, Ehret, and Brick 2022 for a discussion of this).

Second, this may be a result of the CNS and NEP measuring two distinct concepts that can be grouped under the umbrella of environmental orientation. The NEP is explicitly a measure of environmental-attitude while CNS is a measure of emotional connection to nature. For example, a recent study found that harnessing the right mix of emotions is a powerful motivator of climate action (Kovács et al. 2024).

Conclusion

In this paper, we examined the relationship between cultural cognition, environmental orientation, and pro-environmental behavior. We presented evidence that cultural cognition, egalitarianism not communitarianism,

is best thought of as an indirect predictor of pro-environmental behavior. The relationship was mediated by pro-environmental orientation as measured by both the NEP and the CNS.

Because cultural cognition shapes how an individual responds to risk (e.g., Kahan et al. 2012), it was something of a puzzle as to why it only weakly correlates to behavior that protects the environment – an action that can be seen as a way to reduce environmental risk to the individual. The evidence we presented in this paper suggests that environmental orientation – whether attitude (NEP) or emotional connection (CNS) – is a missing link in this relationship.

Further, by explaining behavior with a mediation model we could close some of the so-called attitude-behavior gap. When cultural cognition and environmental orientation are aligned, people are more likely to engage in environmentally protective behavior.

Another implication of this study is further evidence that CNS is a stronger predictor of behavior than NEP. This may be useful for researchers trying to decide which measure of pro-environmental orientation to include in their questionnaire. We would like to see further testing of this finding in studies in other contexts and through meta-analytic techniques.

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